

Geo-Data and Geo-Data Management

Lecture 1: Course Introduction

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About me

Yingjing Huang

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Background: GIScience, GeoAI, Urban Data

More details: <https://yingjinghuang.github.io/>

My research interests

01

GeoAI for urban problems

02

Representation learning of urban data

Example 1 : No “true” greenery

Urban greenery measurement exists limitations



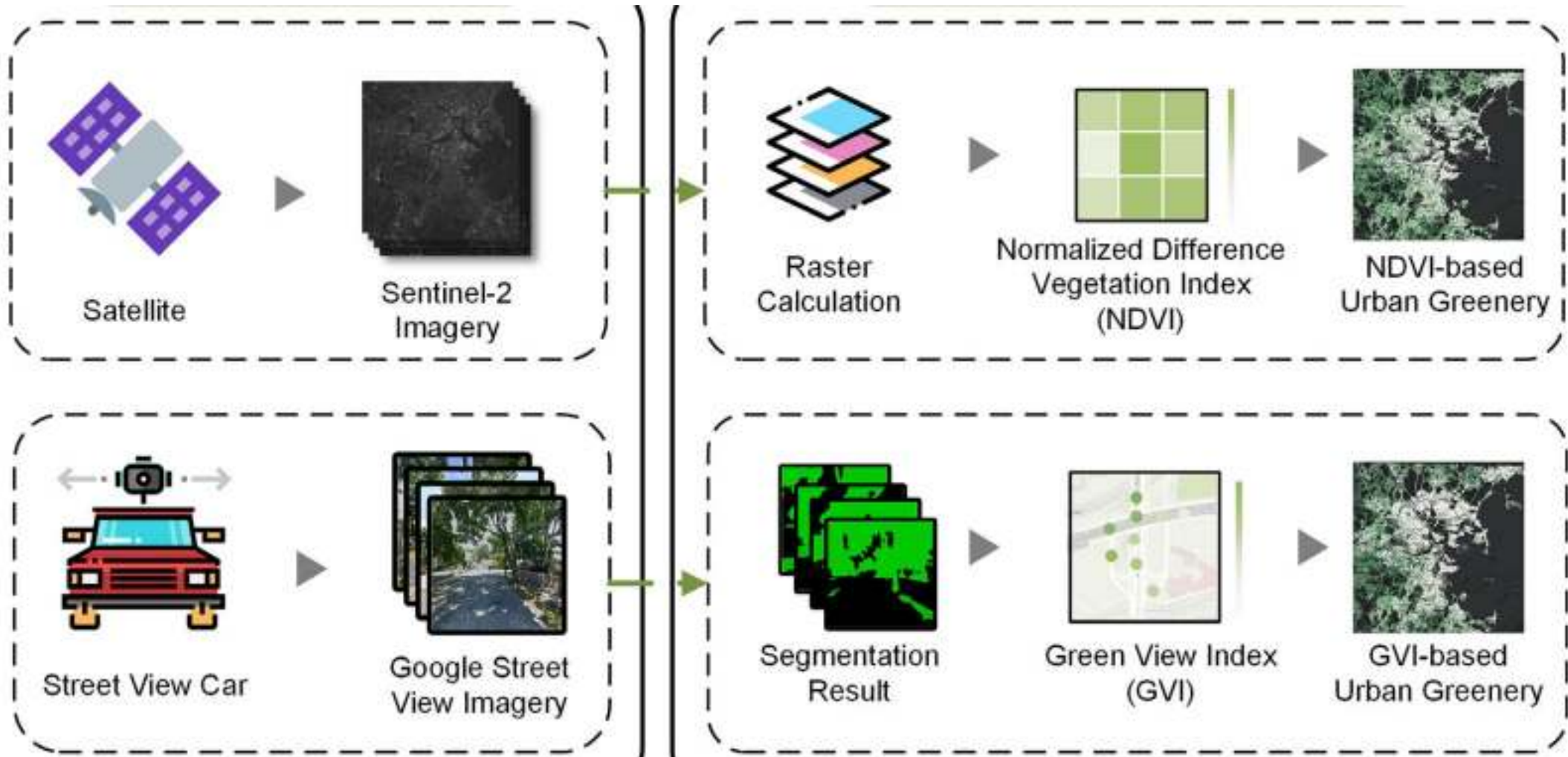
Example 1 : No “true” greenery

Urban greenery measurement exists limitations

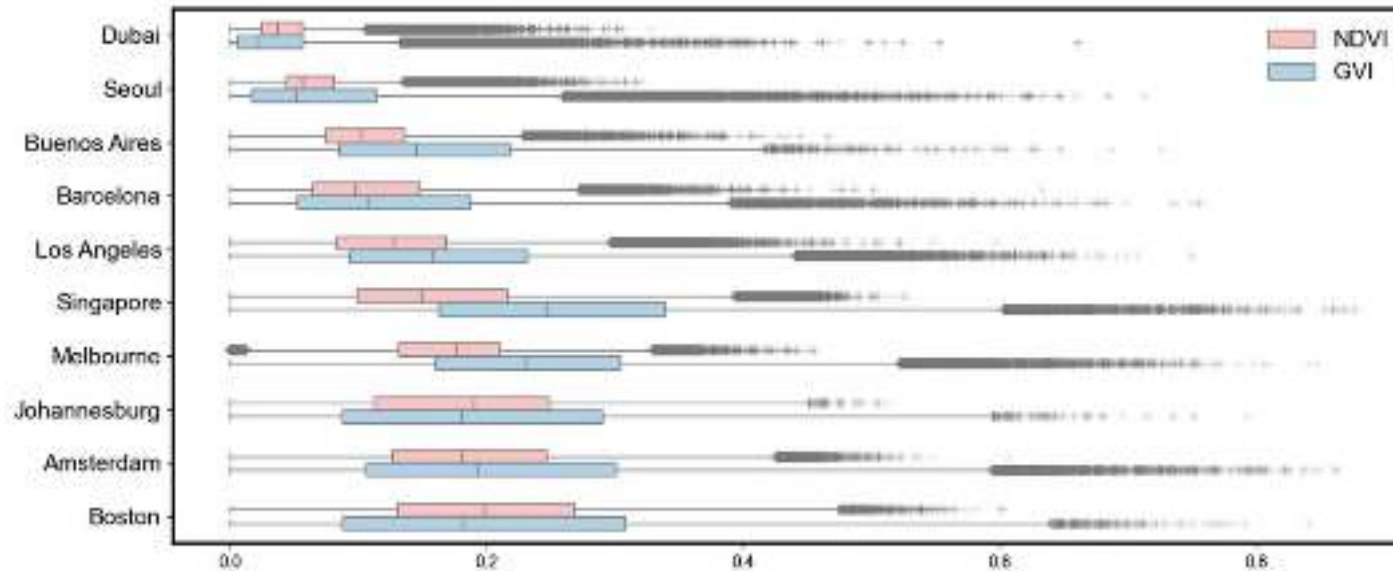


Example 1 : No “true” greenery

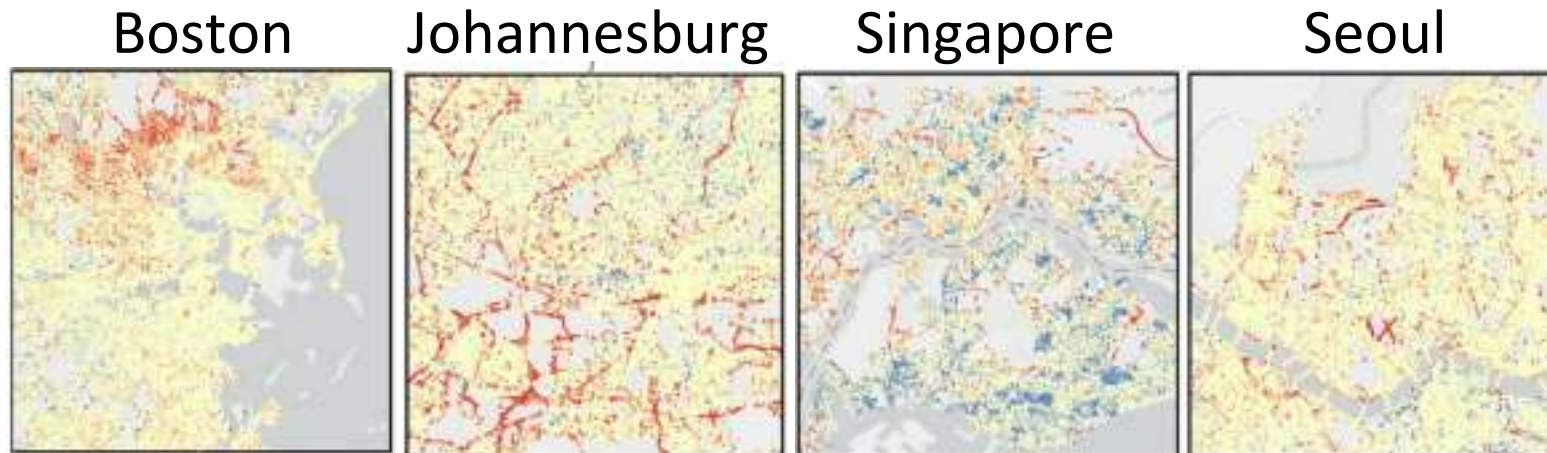
Urban greenery measurement exists limitations



Example 1 : No “true” greenery



Statistical distributions are similar, **but spatial distributions differ!**



Difference between NDVI and GVI



More details: [No “true” greenery paper](#)

Example 2 : Spatiotemporal contrastive learning

Where was this street view image taken?



Example 2 : Spatiotemporal contrastive learning



We design distinct methods to guide computers to focus on the information we need based on different tasks

Who are you?

- Name?
- Area of study?
- Master's program?
- What year are you in the program?
- GIS and/or coding experience?
- What kind of geodata are you familiar with?



Today's plan

1



Course overview

- Course structure
- Grading
- Materials

2



Lecture

- Geodata in the Wild
- How to think about geodata

3



Lab

- Introduction to Jupyter notebook
- Geodata source inventory

Course roadmap

Week	Date	Topic
Week 1	Oct 9	Introduction
Week 2	Oct 16	Vector Data and CRS
Week 3	Oct 23	Raster Data and Georeferencing
Week 4	Nov 13	Vector Data Integration
Week 5	Nov 20	Raster and Vector–Raster Integration
Week 6	Nov 27	Midterm exam (Save the date!)
Week 7	Dec 4	Standards and Interoperability
Week 8	Dec 11	File-based Geodata Storage + Project proposal (Save the date!)
Week 9	Dec 18	Spatial Databases and SQL
Week 10	Jan 8	Reproducible Geodata Management
Week 11	Jan 15	Final project presentation (Save the date!)

No make-up exam!

What is this course about?

When we get a geodata dataset...

- What exactly is it?
- How is location represented?
- Can we combine it with other data?
- How should we store and query it?
- Can someone else understand and reuse it?

How each class works



Lecture

- Main concepts
- Worked examples
- Small methods or code
- ...



Lab

- Walk through part of the notebook
- Understand how the code works
- Practice by yourself
- ...

A lab checkpoint should be submitted every week

Grading

- **Participation (10%)**
 - Submit a lab checkpoint for each class **by 23:59 on the day of the class**
- **Midterm exam (40%)**
 - Held in Week 6
- **Project proposal (20%)**
 - Detailed instructions will be released in Week 8
 - The proposal should be submitted by each team in Week 9
- **Final project (30%)**
 - Team presentation in Week 11

Course materials

GitHub



- Teaching materials
- Public access

<https://yingjinghuang.github.io/Geo-Data-and-Geo-Data-Management/>

Moodle

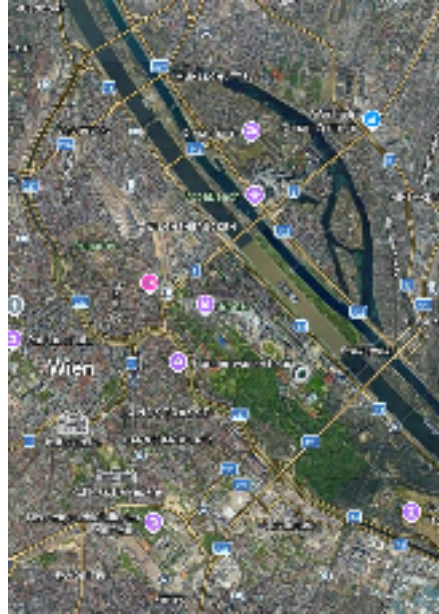


- Teaching materials
- Announcements
- Submissions (e.g., lab checkpoints)
- Grades

Geodata in the Wild



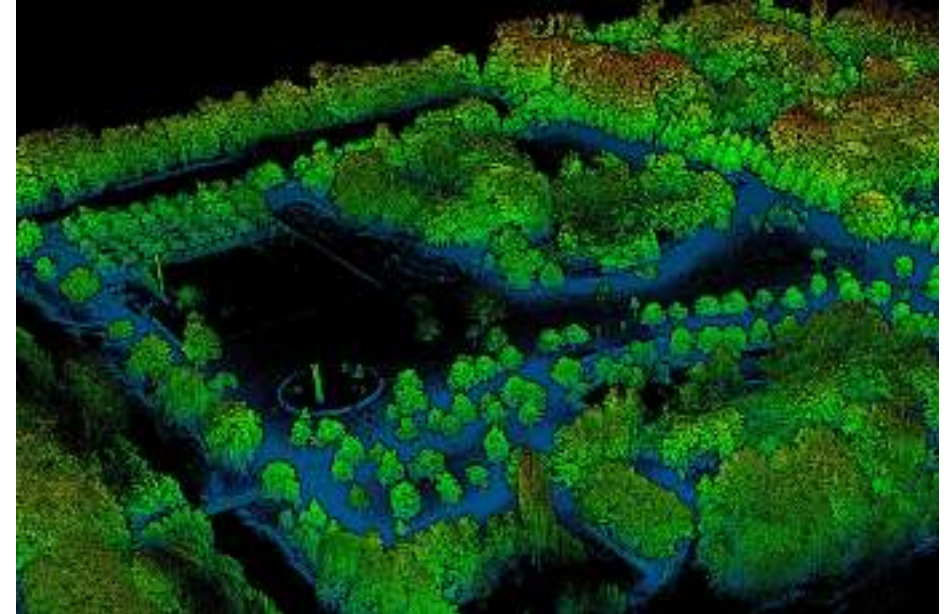
Digital map
(vector data)



Satellite image
(raster data)



GPS trace
(trajectory data)



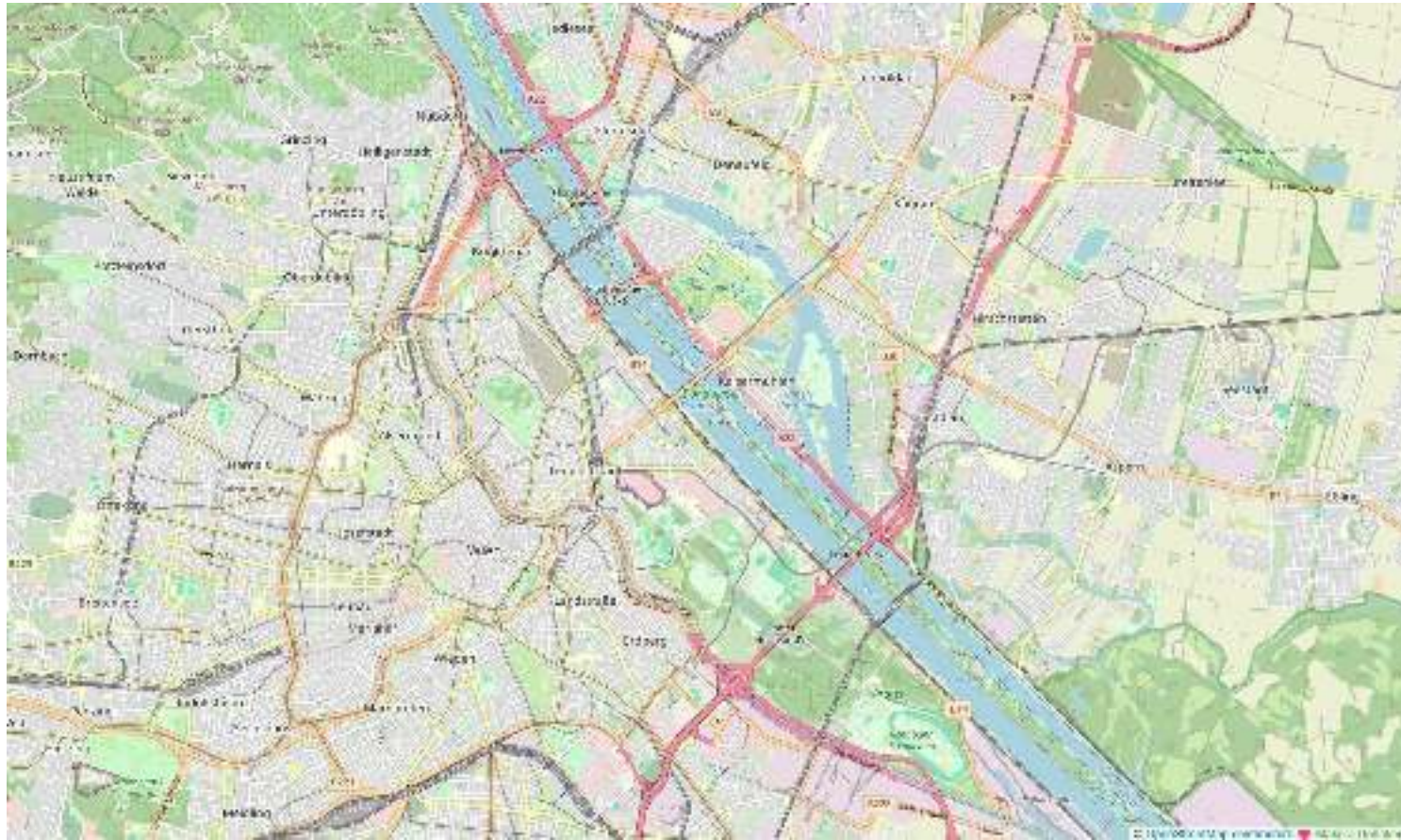
Point cloud
(3D data)

Are all of these geodata?

Geodata example: OpenStreetMap



www.openstreetmap.org



What we see

- a rendered map

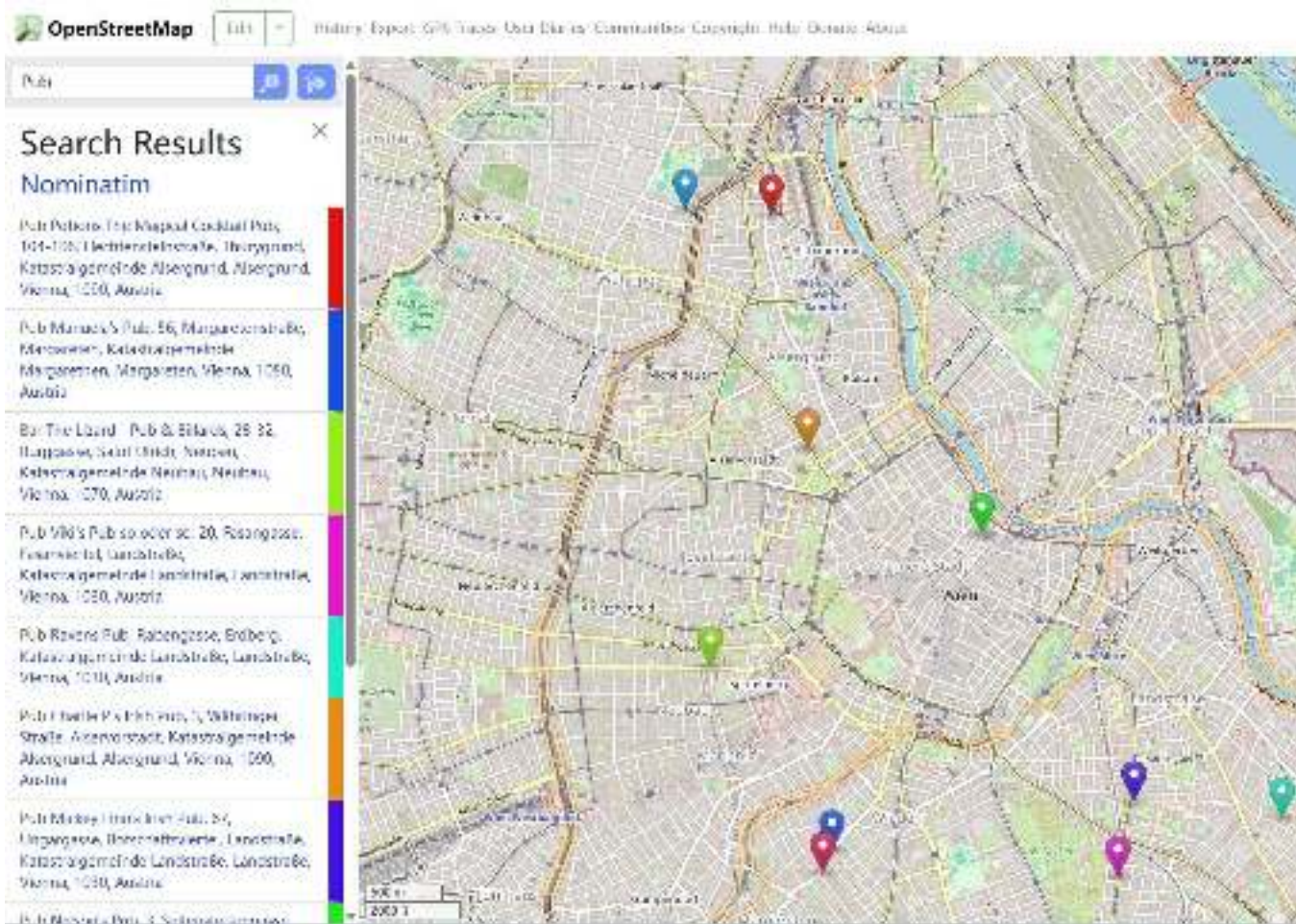
What exists behind it

- places
- roads
- buildings / areas
- attributes / tags
- relationships
- edits over time



The map is one view of the underlying geodata.

OpenStreetMap: Point Data



Pubs in Vienna

For example:

POI (Points of Interest)

What a point gives us:

- a specific location (latitude and longitude)
- can carry attributes (e.g. pub, bus stop)

OpenStreetMap: Line Data



- Road (highway)
- Railway
- River / Waterway
- - - Path / Cycleway
- Other linear features

Zoomed-in examples

Road network



Railway network



River and canal



Cycleways and paths



OpenStreetMap: Polygon Data

Examples from OpenStreetMap



● **Buildings**
building = yes



● **Parks and Green Spaces**
leisure = park



● **Residential Areas**
landuse = residential



● **Commercial Areas**
landuse = commercial



● **Industrial Areas**
landuse = industrial



● **Water Bodies**
natural = water

For example:

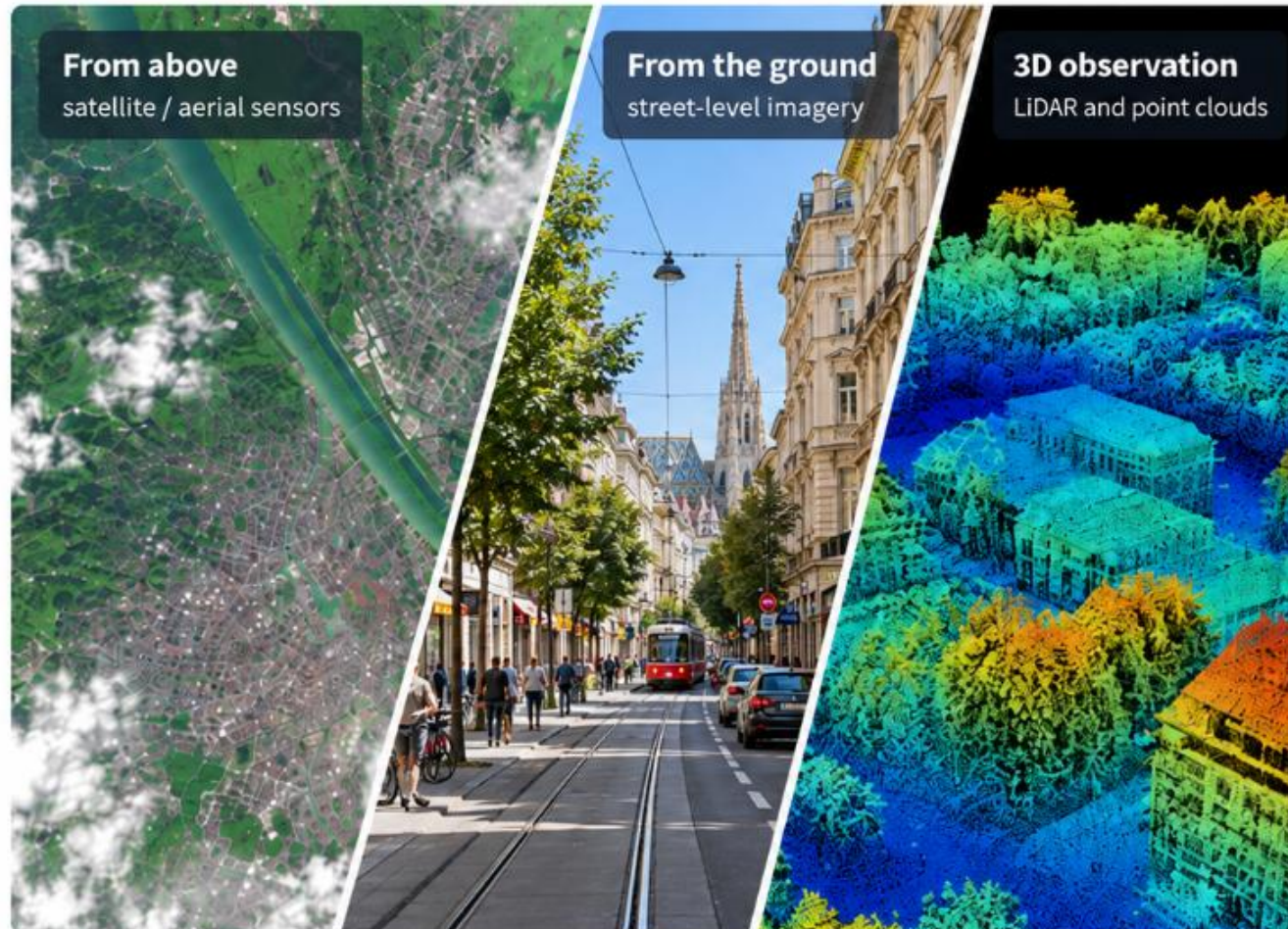
AOI (Areas of Interest)

Building footprints

Land use

Geodata Example: Earth Observation

use cameras or sensors to observe the physical environment

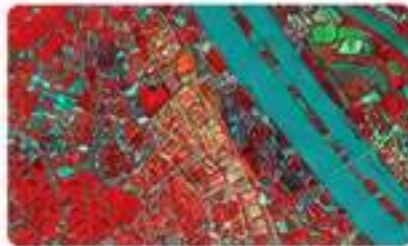


Earth observation: Remote sensing imagery

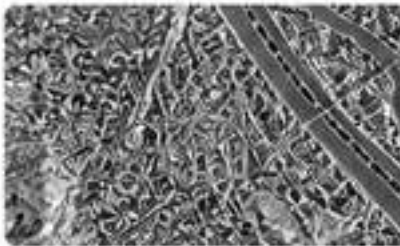
Examples of remote sensing imagery



Optical Imagery (True Color)
natural color view
(e.g., Sentinel-2, Landsat)



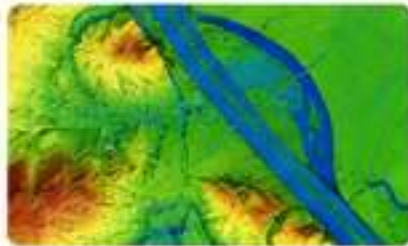
False-Color Imagery
highlights vegetation
and land cover (e.g., NDVI)



Radar Imagery (SAR)
works in clouds and
at night (e.g., Sentinel-1)



Thermal Imagery
surface temperature
(e.g., Landsat thermal)



Elevation Models (DEM)
surface height (e.g., SRTM,
Copernicus DEM)



Very High Resolution Imagery
detailed view (e.g., WorldView,
Pleiades)

Observe the Earth's surface from above

Data form: imagery / gridded pixels

- Each pixel stores one or more measured values
- Images may contain multiple spectral bands

Key characteristics:

- Sensors differ in **spatial, spectral, and temporal resolution**
- Can repeatedly observe large areas over time

Earth observation: Street-level imagery



Sources:

Google Street View, Baidu Street View, Mapillary (**Open source!**), KartaView (**Open source!**), ...

Observe the environment from a ground-level perspective

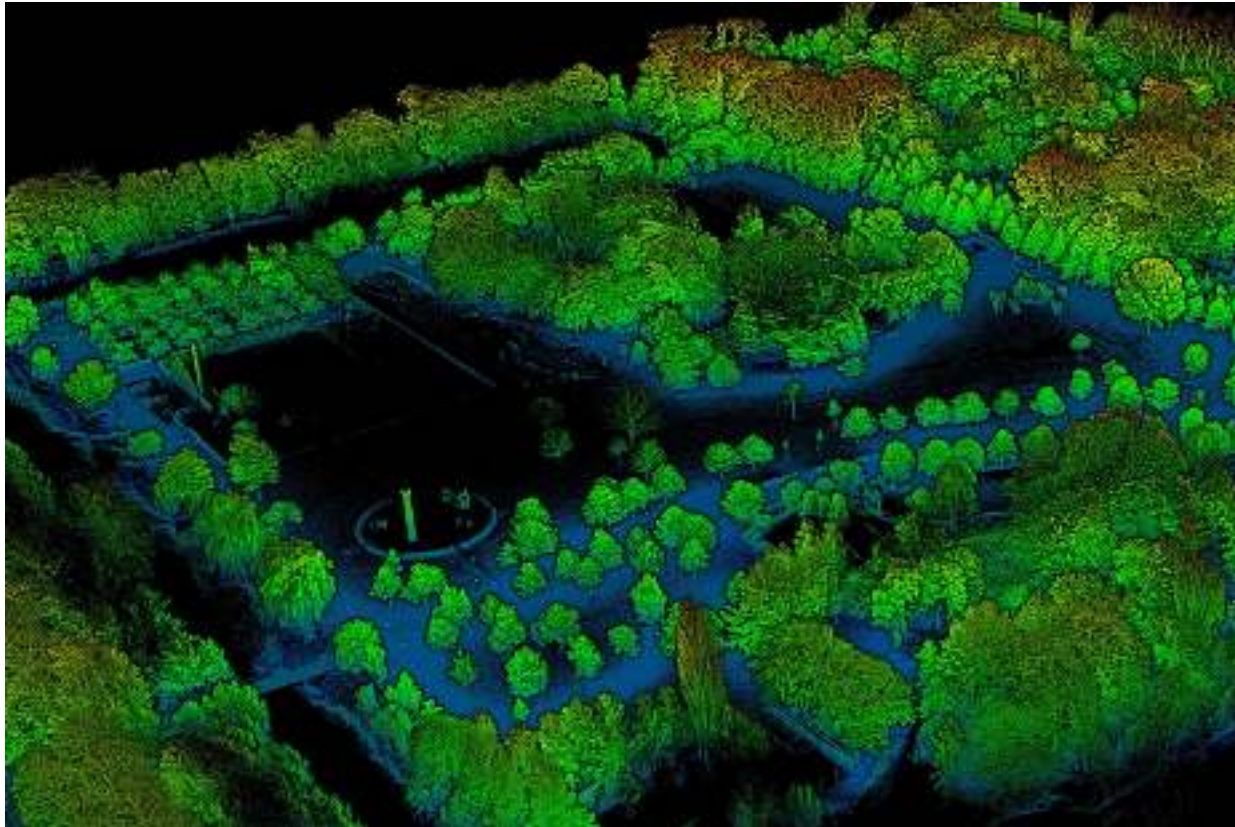
Data form: imagery / panoramic imagery

- Usually linked to a geographic location
- May include capture time and viewing direction

Key characteristics:

- Captures fine visual detail along streets and routes
- Often collected as a sequence of georeferenced images

Earth observation: LiDAR and Point Clouds



Measure 3D structure of the environment using laser pulse

Data form: 3D point clouds

Each point has 3D coordinates (x,y,z)

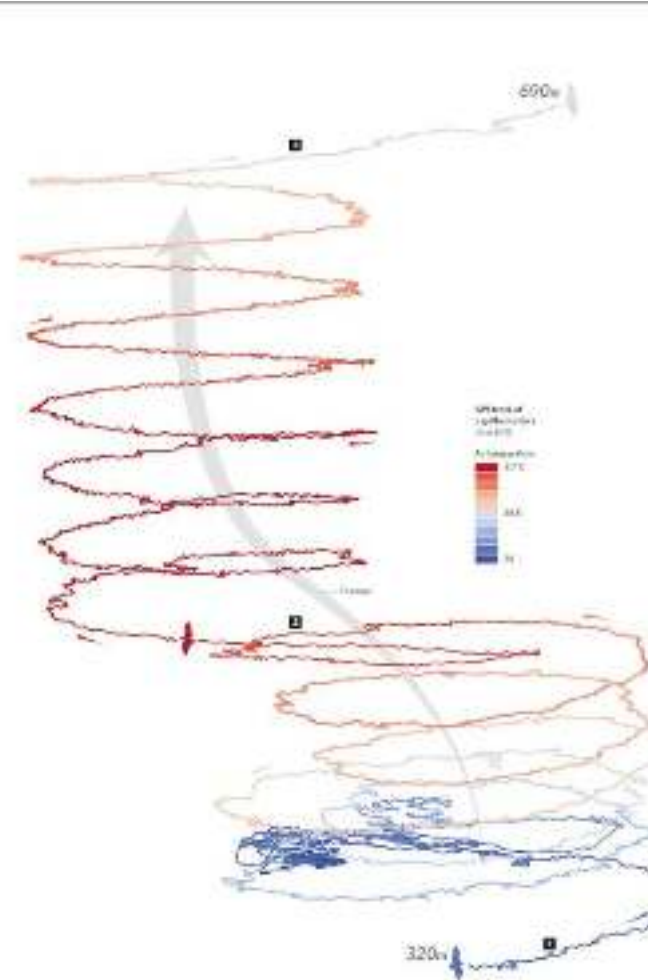
Sources:

- Airborne LiDAR (e.g., national mapping agencies, OpenTopography)
- Terrestrial or mobile LiDAR (e.g., vehicles, handheld devices)

Geodata Example: GPS and Trajectories



Wildlife tracking maps



Tracks the movement of people, vehicles, or other objects over time

Data form: spatiotemporal points

A sequence of geographic locations with timestamps

Sources:

- GNSS devices in smartphones or dedicated trackers
- Public transport data (e.g., bus/train GPS records)
- Crowdsourced or commercial mobility datasets (e.g., phone location data)

Geodata Example: Vienna Open Data

Many cities have their own geo database

Themen



Internationale Themen



Landwirtschaft, Fischerei,
Forstwirtschaft und
Nahrungsmittel



Bevölkerung und Gesellschaft



Gesundheit



Verkehr



Bildung, Kultur und Sport



Justiz, Rechtssystem und
öffentliche Sicherheit



Wissenschaft und Technologie



Regionen und Städte



Energie



Regierung und öffentlicher
Sektor



Umwelt



Wirtschaft und Finanzen

www.data.gv.at

What counts as geodata?

————— They all relate to geographic information —————

Geodata is a broad family of data that **carries geographic meaning**

Understanding Geodata

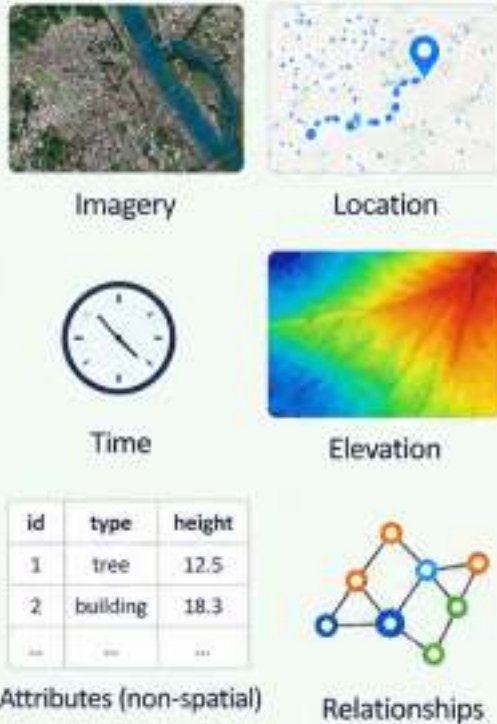
1 Source

Where does the data come from?



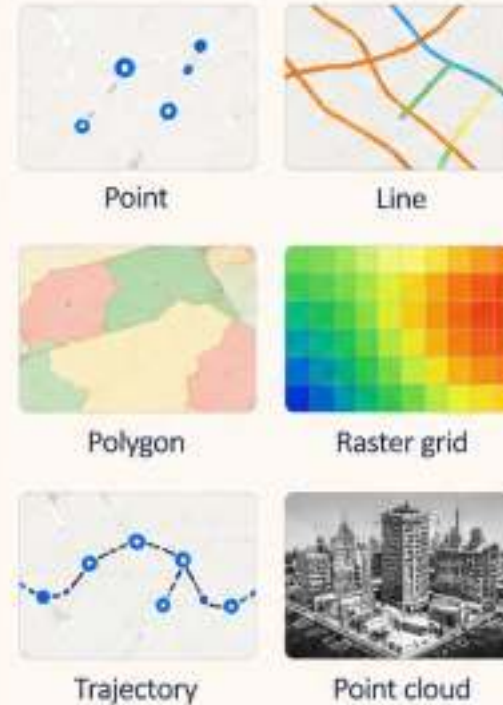
2 Information Type

What kind of information is recorded?



3 Representation

How is the information organised to represent geographic meaning?

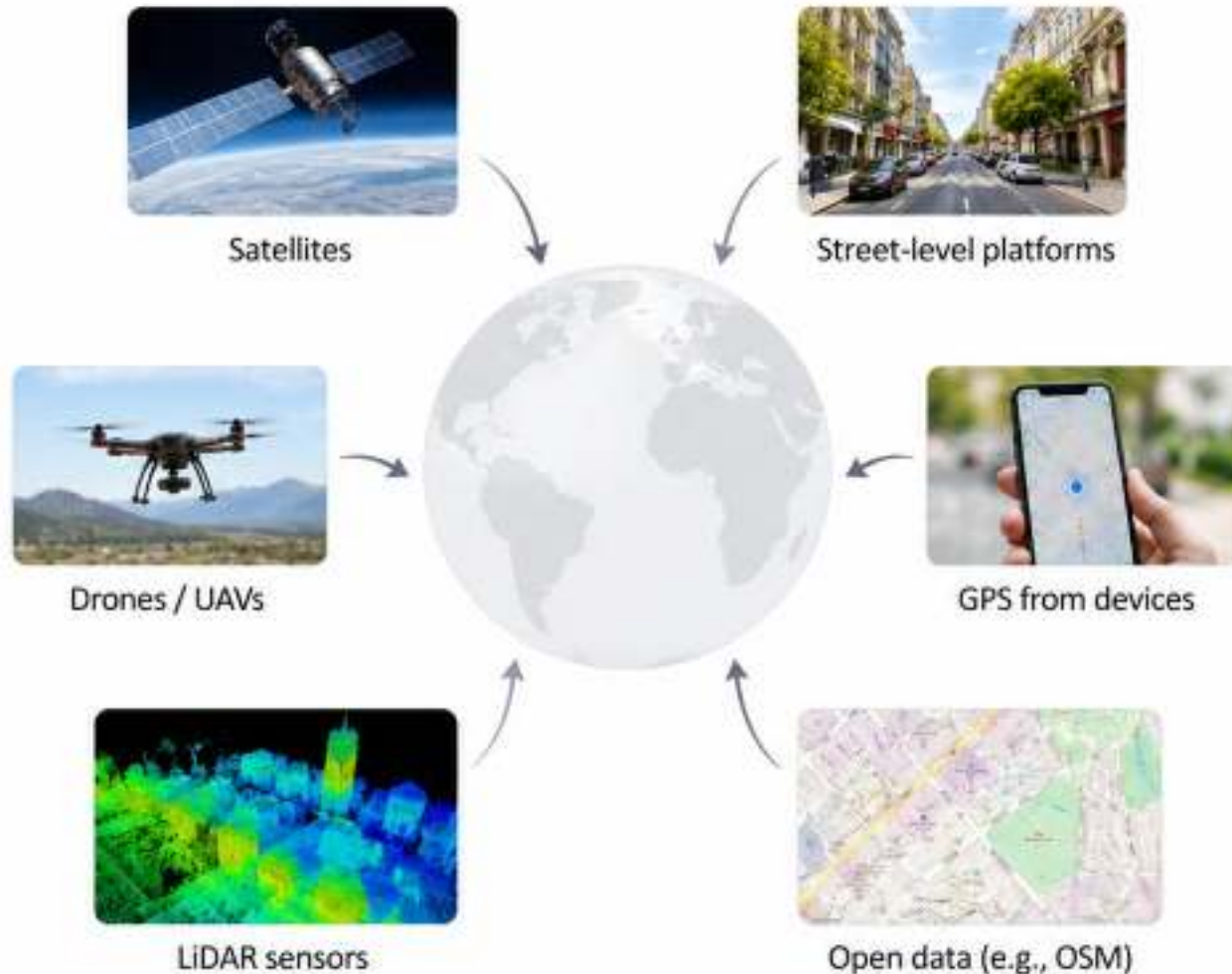


4 Storage and Access

How is the data stored or accessed?



Understanding Geodata: Source



1

Source

Where does the data come from?



Remote sensing satellites

e.g., Sentinel-2, Landsat



Street-level platforms

e.g., Google Street View, Mapillary



Drones / UAVs

e.g., aerial imagery, LiDAR



GPS from devices

e.g., smartphones, vehicles



Open data providers

e.g., OpenStreetMap, city open data portals



Others

e.g., surveys, crowdsourcing, institutional data

Understanding Geodata: Information type

Imagery

pixels / images



Location

coordinates / positions



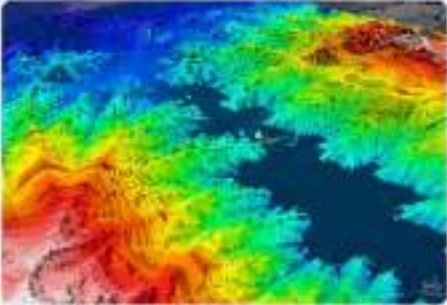
Time

timestamps / sequences



Elevation / range

height / distance measurements



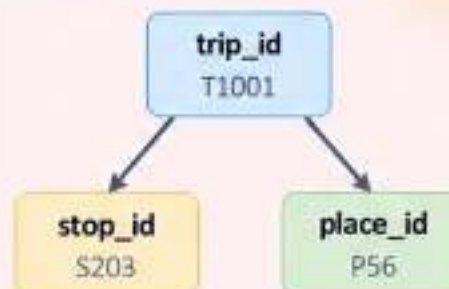
Attributes / text

names / categories / descriptions

name	Central Park
type	Park
category	Recreation
area_km2	3.41
description	Urban park in NYC

Relationships / IDs

links between records



2

Information Type

What information is recorded?



Imagery

e.g., satellite photos, street photos



Location

e.g., coordinates, GPS positions



Time

e.g., timestamps, time series



Elevation / range

e.g., LIDAR measurements, DEM values



Attributes / text

e.g., names, categories, descriptions



Relationships / IDs

e.g., linked records, trip_id, stop_id

Understanding Geodata: Representation

Point

Single locations



Line

Linear features



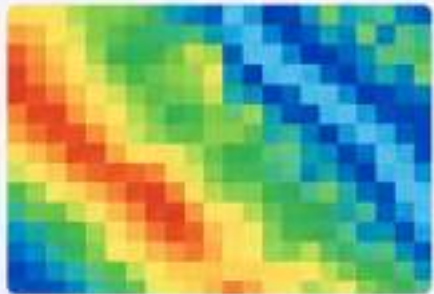
Polygon

Areas / boundaries



Raster grid

Pixels / cells



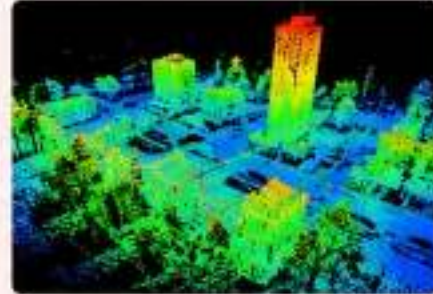
Trajectory

Ordered locations over time



Point cloud

3D points



3

Representation

How is geographic meaning encoded?



Point

e.g., pubs, bus stops



Line

e.g., roads, rivers



Polygon

e.g., buildings, land use



Raster grid

e.g., satellite imagery, DEM



Trajectory

e.g., GPS tracks



Point cloud

e.g., LIDAR



Others

e.g., networks, graphs

Understanding Geodata: Storage and access

Files

saved as files



GeoTIFF

GeoJSON

LAS

Databases

stored in tables / systems



PostGIS, spatial DB

id	name	geom
1	Park	POINT ...
2	River	LINE ...
3	Building	POLYGON ...
...	...	...

Web services / APIs

requested online



Request

Response



REST API, WMS / tiles

Cloud platforms / portals

accessed through platforms



Google Earth Engine, open data portal

4

Storage and Access

How is it stored or accessed?



Files

e.g., GeoTIFF, GeoJSON, LAS



Databases

e.g., PostGIS, spatial DB



Web services / APIs

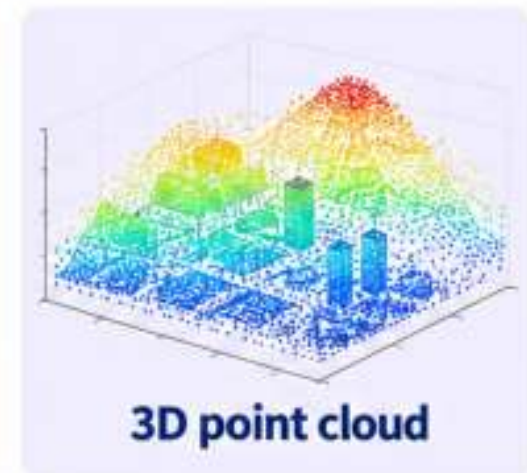
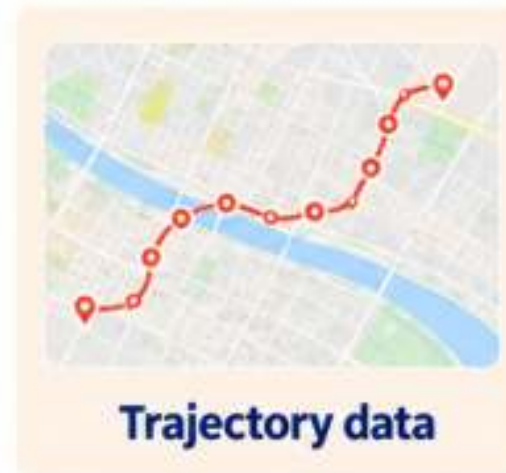
e.g., REST API, WMS / tiles



Cloud platforms / portals

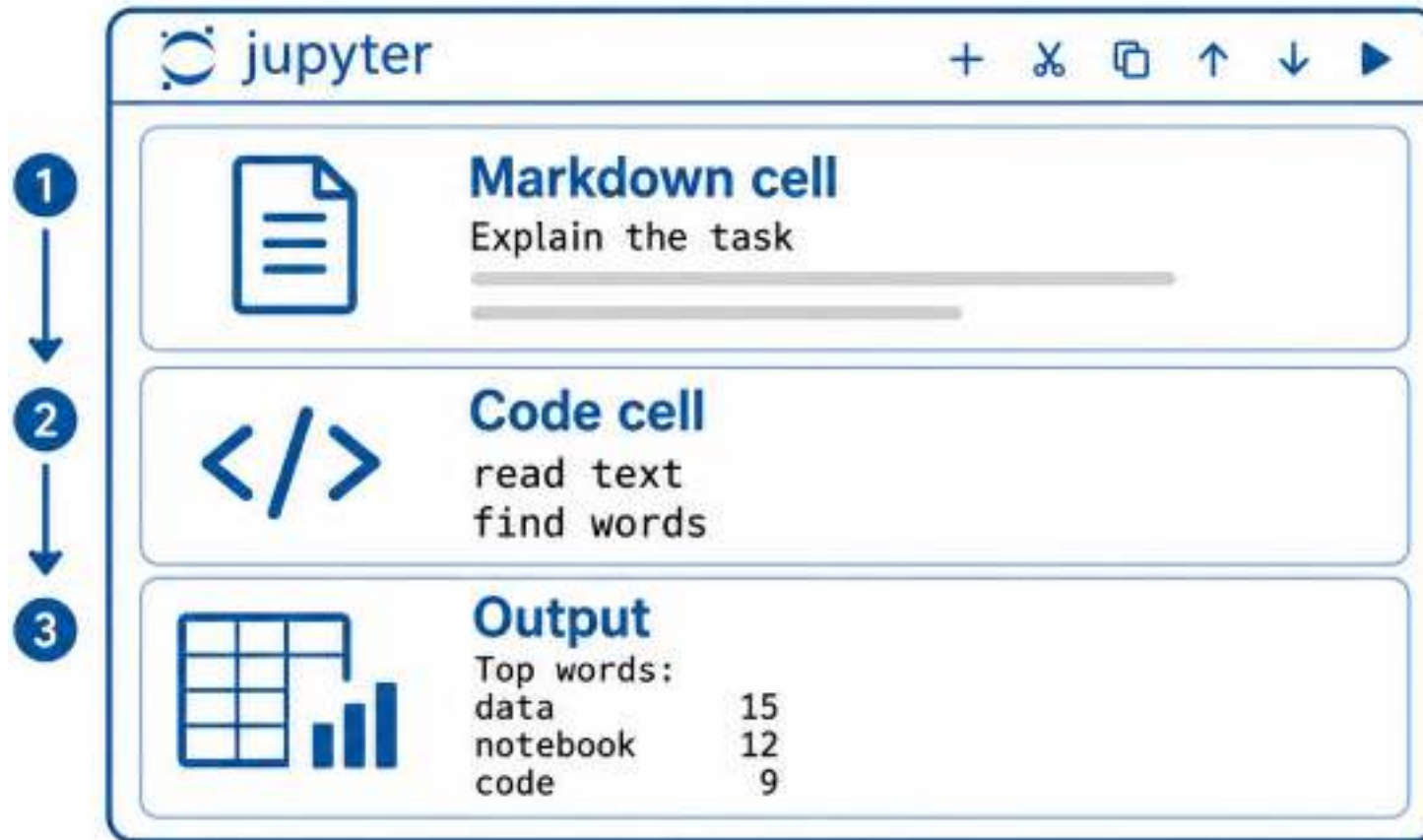
e.g., Google Earth Engine, open data portal

First lab: Acquire, load, and visualize geodata

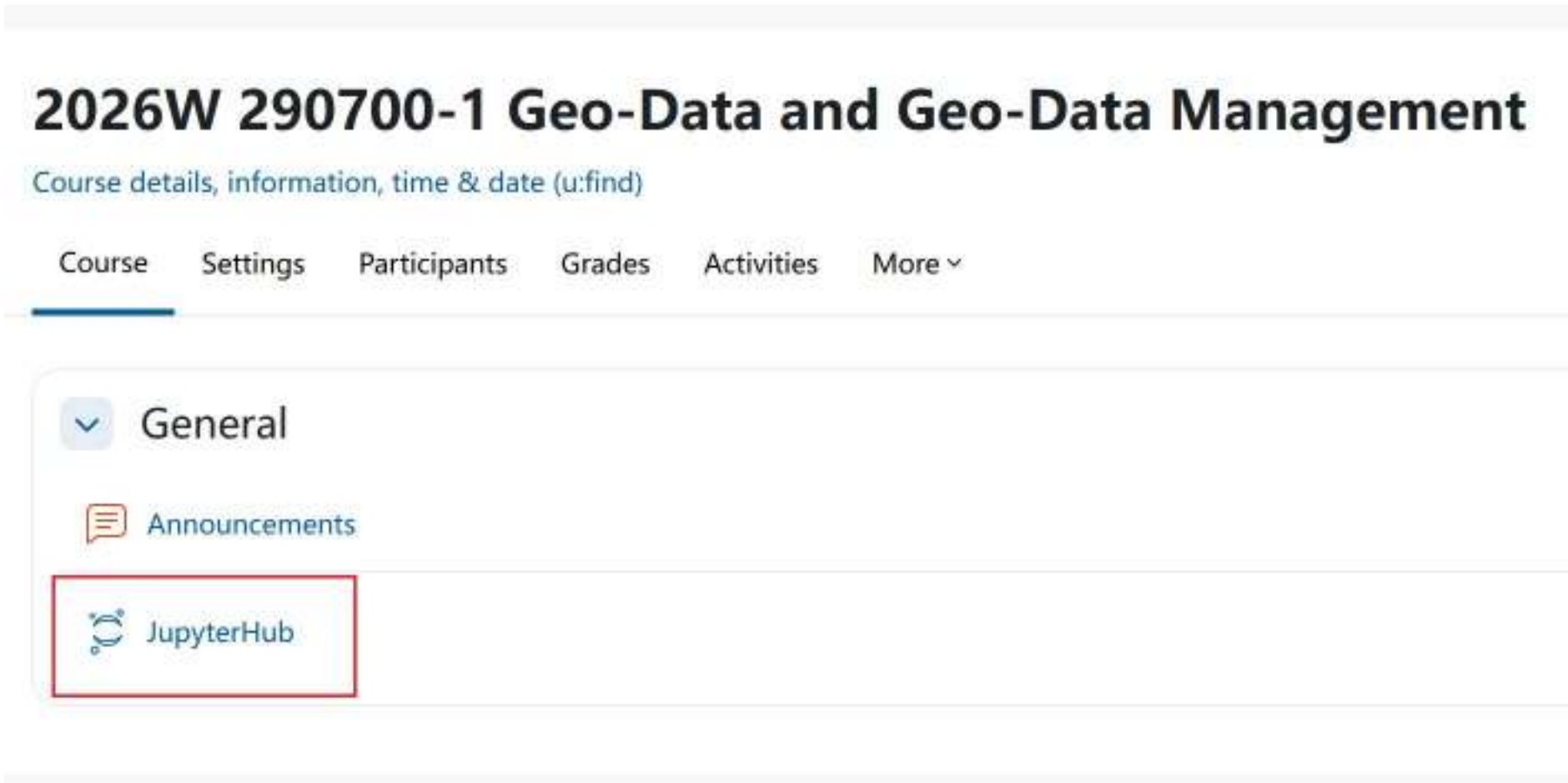


What is a Jupyter notebook?

A notebook keeps **explanation**, **code**, and **results** together



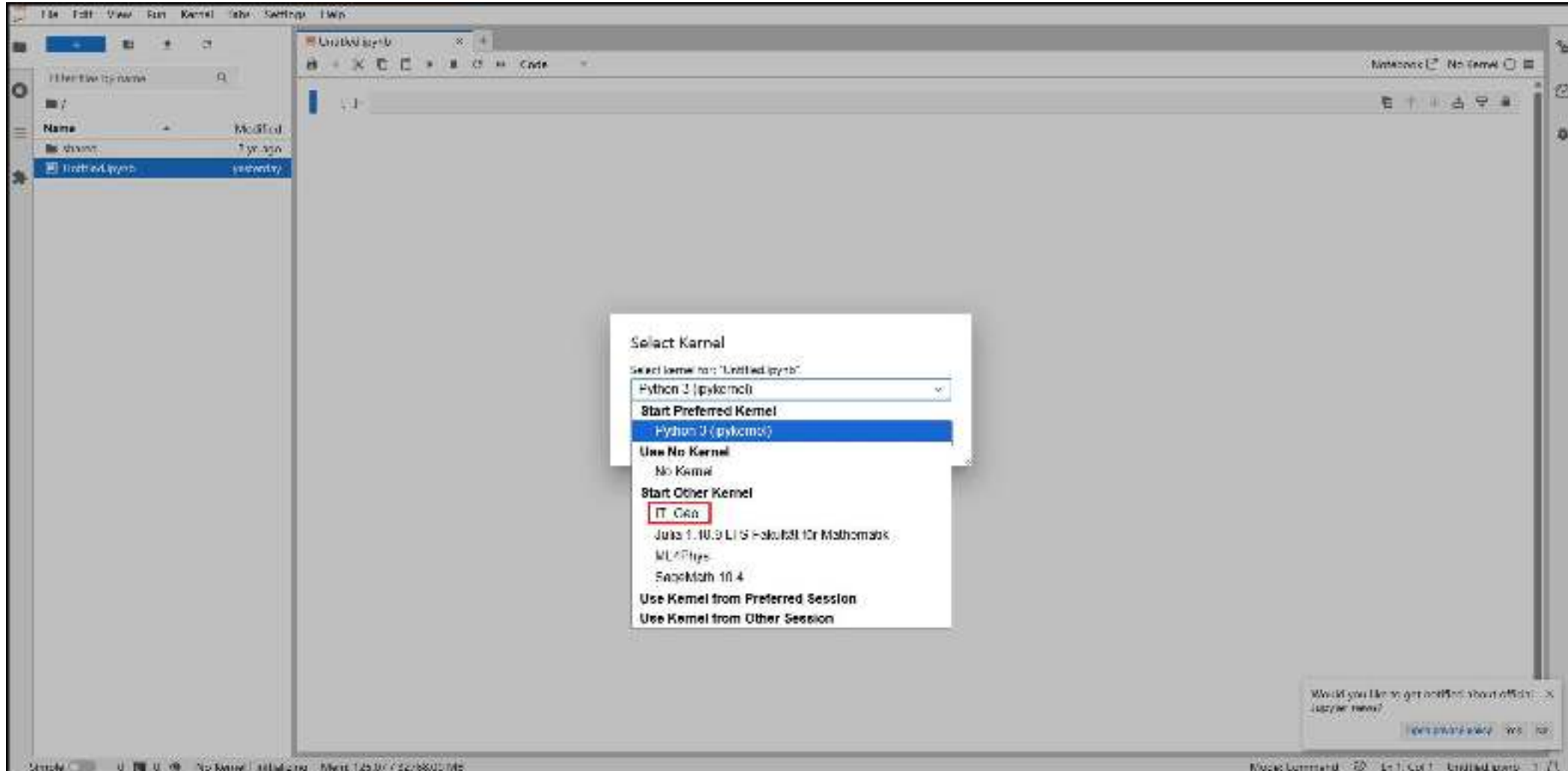
How to use notebooks in this course



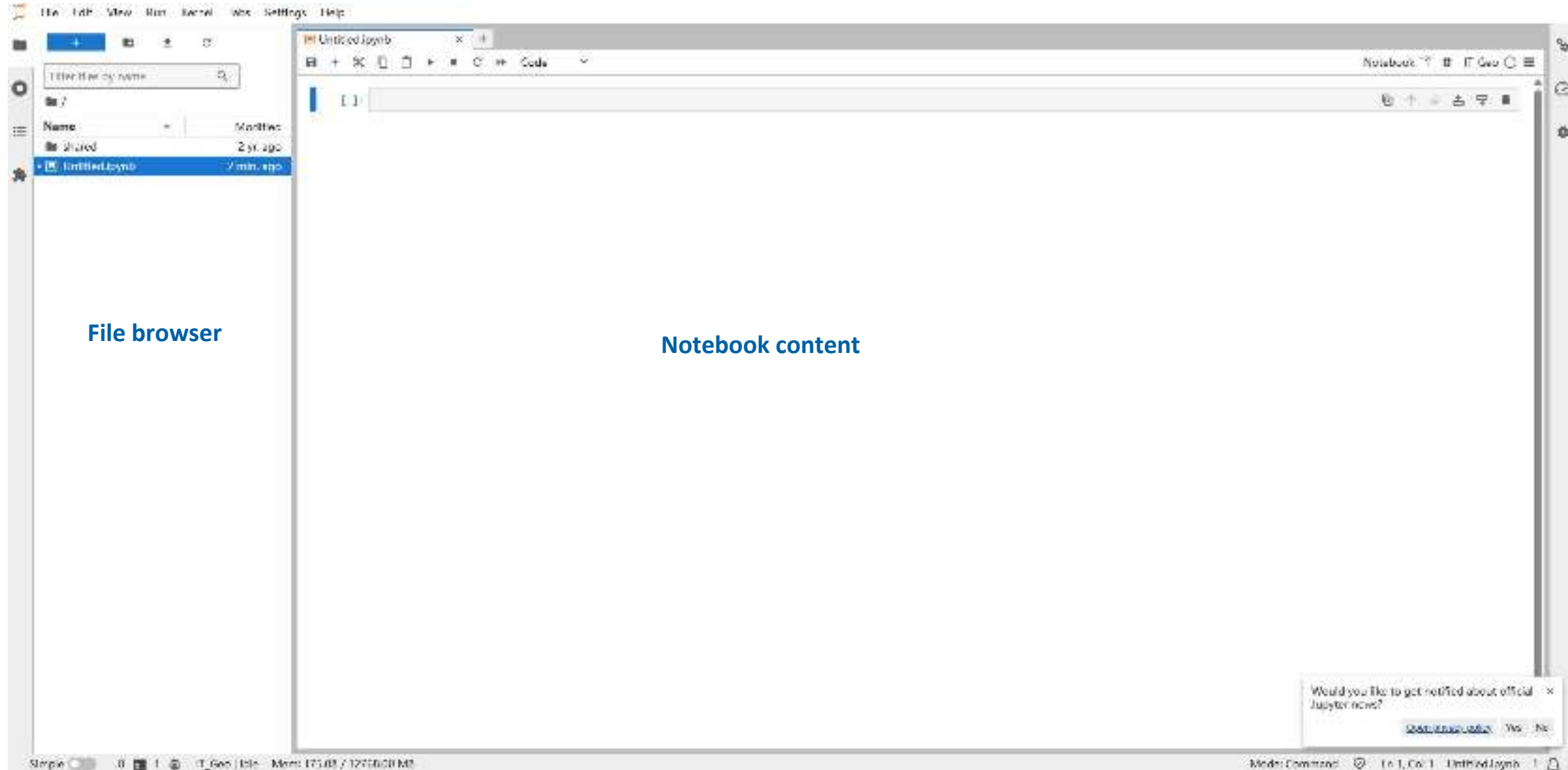
How to use notebooks in this course



How to use notebooks in this course



How to use notebooks in this course



Lab time

*Please remember to submit your lab checkpoint
by the end of the day.*